

CICD Pipeline MAVEN JENKINS and TOMCAT

Create instances

☐	Name 	Instance ID	Instance state 	Instance type 	Status check	Alarm status	Availability Zone 
☐	jenkins-server	i-044ba16879838dece	 Running  	t3.micro	 3/3 checks passed	View alarms +	ap-south-2a
☐	tomcat-server	i-0f641311bf71b2e91	 Running  	t3.micro	 3/3 checks passed	View alarms +	ap-south-2a

Connect the instance using SSH

`ssh -i "jenkins-fc-training.pem" ec2-user@ec2-16-112-64-77.ap-south-2.compute.amazonaws.com`

Jenkins Installation

What is Jenkins?

Jenkins is an open-source automation server written in Java. It facilitates continuous integration and continuous deployment (CI/CD) by automating the building, testing, and deployment of software projects. Jenkins is highly extensible, with a vast ecosystem of plugins that allow it to integrate with various tools and services, making it a popular choice for DevOps teams.

Key Features

- **Continuous Integration/Continuous Deployment (CI/CD):** Automates the process of integrating code changes, running tests, and deploying applications.
- **Extensive Plugin Ecosystem:** Over 1,800 plugins to integrate with tools like Git, Docker, Kubernetes, AWS, and more.
- **Pipeline Support:** Define build processes as code using Jenkins Pipeline, enabling complex workflows.
- **Distributed Builds:** Supports master-agent architecture to distribute workloads across multiple machines.
- **Customizable:** Highly configurable to suit various project needs, from simple builds to complex deployment pipelines.
- **Community Support:** Backed by a large community, providing extensive documentation and resources.

Pre-Requisites

To install Jenkins on a Debian-based system (e.g., Ubuntu), ensure the following requirements are met:

- **Operating System:** Debian-based Linux distribution (e.g., Ubuntu 20.04 or later).
- **Java (JDK):** Jenkins requires Java to run. OpenJDK 17 is recommended.
- **Hardware:** At least 1 GB of RAM and sufficient disk space for builds and artifacts.
- **Network:** Internet access for downloading Jenkins and plugins; port 8080 open for accessing the Jenkins web interface.

- **User Permissions:** Administrative (sudo) access to install packages.

Installation Steps

```
mount -o remount,size=4G /tmp
```

```
sudo yum update -y
```

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \
```

```
https://pkg.jenkins.io/rpm-stable/jenkins.repo
```

```
sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins.io-2026.key
```

```
sudo yum upgrade -y
```

```
sudo yum install java-21-amazon-corretto -y
```

```
sudo yum install jenkins -y
```

```
sudo systemctl enable jenkins
```

```
sudo systemctl start jenkins
```

```
sudo systemctl status jenkins
```

```
sudo yum install git maven -y
```

```
Installed:
jenkins-2.555.1-1.noarch

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-04-19 06:10:32 UTC; 127ms ago
     Main PID: 24557 (java)
        Tasks: 40 (limit: 1014)
       Memory: 371.3M
          CPU: 19.660s
      CGroup: /system.slice/jenkins.service
              └─24557 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Apr 19 06:10:25 ip-172-31-1-127.ap-south-2.compute.internal jenkins[24557]: [LF]> 8f5267cc60e74be09be5c50655130796
Apr 19 06:10:25 ip-172-31-1-127.ap-south-2.compute.internal jenkins[24557]: [LF]>
Apr 19 06:10:25 ip-172-31-1-127.ap-south-2.compute.internal jenkins[24557]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Apr 19 06:10:25 ip-172-31-1-127.ap-south-2.compute.internal jenkins[24557]: [LF]>
Apr 19 06:10:25 ip-172-31-1-127.ap-south-2.compute.internal jenkins[24557]: [LF]> *****
Apr 19 06:10:25 ip-172-31-1-127.ap-south-2.compute.internal jenkins[24557]: [LF]> *****
Apr 19 06:10:25 ip-172-31-1-127.ap-south-2.compute.internal jenkins[24557]: [LF]> *****
Apr 19 06:10:32 ip-172-31-1-127.ap-south-2.compute.internal jenkins[24557]: 2026-04-19 06:10:32.304+0000 [id=31] INFO jenkins.Init
Apr 19 06:10:32 ip-172-31-1-127.ap-south-2.compute.internal jenkins[24557]: 2026-04-19 06:10:32.339+0000 [id=24] INFO hudson.lifecycle
Apr 19 06:10:32 ip-172-31-1-127.ap-south-2.compute.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.

lines 1-20/20 (END)
```

Access the jenkins through web browser using Ipaddress with port 8080

Get the password using cat command

```
cat /var/lib/jenkins/secrets/initialAdminPassword
```

You can view the below snap

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

Click on install suggested plugins → it will install the dependencies

After install you can see the below snap to provide user details

Create First Admin User

Username

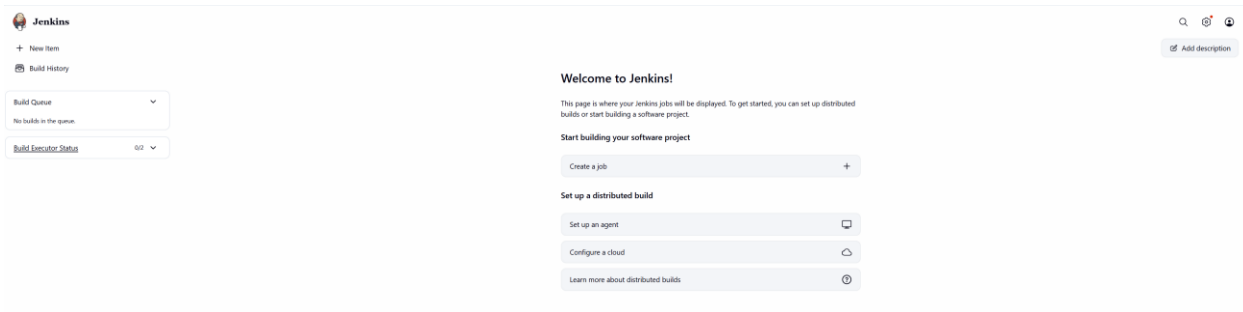
Password

Confirm password

Full name

E-mail address

Provide your details and click on save and continue → click on finish button to view jenkins GUI



Now connect to tomcat server using SSH

`ssh -i "jenkins-fc-training.pem" ec2-user@ec2-98-130-124-2.ap-south-2.compute.amazonaws.com`

Once connected follow the below official document to download apache package

Work on /opt directory

`cd /opt`

install the java package

`sudo yum install java-21-amazon-corretto -y`

`wget https://dldcn.apache.org/tomcat/tomcat-9/v9.0.117/bin/apache-tomcat-9.0.117.tar.gz`

Once downloaded extract the package

`tar -xvf apache-tomcat-9.0.117.tar.gz`

Once extracted the packages go to the apache directory

`cd apache-tomcat-9.0.117`

go to the below directory, you can view the context.xml

`cd /opt/apache-tomcat-9.0.117/webapps/manager/META-INF`

`vi context.xml`

inside context you need to remove lines in between the below

`<Context antiResourceLocking="false" privileged="true" >`

Remove what ever the lines you have

`</Context>`

`:wq (save and exit)`

Now go to conf directory

```
[root@tomcat META-INF]# cd ../../
[root@tomcat webapps]# cd ../conf/
[root@tomcat conf]# ll
total 228
-rw-----. 1 root root 12953 Mar 30 18:21 catalina.policy
-rw-----. 1 root root 7654 Mar 30 18:21 catalina.properties
-rw-----. 1 root root 1400 Mar 30 18:21 context.xml
-rw-----. 1 root root 1149 Mar 30 18:21 jaspic-providers.xml
-rw-----. 1 root root 2313 Mar 30 18:21 jaspic-providers.xsd
-rw-----. 1 root root 4003 Mar 30 18:21 logging.properties
-rw-----. 1 root root 8022 Mar 30 18:21 server.xml
-rw-----. 1 root root 2756 Mar 30 18:21 tomcat-users.xml
-rw-----. 1 root root 2558 Mar 30 18:21 tomcat-users.xsd
-rw-----. 1 root root 174047 Mar 30 18:21 web.xml
[root@tomcat conf]# vi server.xml
```

Change the connector port 8080 to 8081 (or anyother port like 8082 8083 or 8090 etc)

Now edit the tomcat-users.xml using vi editor

vi tomcat-users.xml

esc 100dd to delete all the content

copy and paste the below

```
<?xml version="1.0" encoding="utf-8"?>
<tomcat-users>
<role rolename="manager-gui"/>
<user username="tomcat" password="tomcat" roles="manager-gui, manager-script, manager-status"/>
</tomcat-users>
```

Now come to bin directory

cd ../bin/

start the tomcat server using the command below

```
[root@tomcat bin]# ./startup.sh
Using CATALINA_BASE: /opt/apache-tomcat-9.0.117
Using CATALINA_HOME: /opt/apache-tomcat-9.0.117
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.117/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/apache-tomcat-9.0.117/bin/bootstrap.jar:/opt/apache-tomcat-9.0.117/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

Access through web browser using ipaddress with port 8081

[Home](#) [Documentation](#) [Configuration](#) [Examples](#) [Wiki](#) [Mailing Lists](#) [Find Help](#)

Apache Tomcat/9.0.117



If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

[Security Considerations How-To](#)

[Manager Application How-To](#)

[Clustering/Session Replication How-To](#)

Server Status

Manager App

Host Manager

Developer Quick Start

[Tomcat Setup](#)
[First Web Application](#)

[Realms & AAA](#)
[JDBC DataSources](#)

[Examples](#)

[Servlet Specifications](#)
[Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 9.0 access to the manager application is split between different users.
[Read more...](#)

[Release Notes](#)
[Changelog](#)
[Migration Guide](#)
[Security Notices](#)

Documentation

[Tomcat 9.0 Documentation](#)
[Tomcat 9.0 Configuration](#)
[Tomcat Wiki](#)

Find additional important configuration information in:

`$CATALINA_HOME/RUNNING.txt`

Developers may be interested in:

[Tomcat 9.0 Bug Database](#)
[Tomcat 9.0 JavaDocs](#)
[Tomcat 9.0 Git Repository at GitHub](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

[tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).

[tomcat-users](#)
User support and discussion

[taglibs-user](#)
User support and discussion for [Apache Taglibs](#)

[tomcat-dev](#)
Development mailing list, including commit messages

Other Downloads

[Tomcat Connectors](#)
[Tomcat Native](#)
[Taglibs](#)
[Deployer](#)

Other Documentation

[Tomcat Connectors](#)
[mod_jk Documentation](#)
[Tomcat Native](#)
[Deployer](#)

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[Overview](#)
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Click on manager app it will prompt username and password

Insert your username and password details

Once logged in you can view the below GUI





Tomcat Web Application Manager

Message: OK

Manager

HTML Manager Help

Manager Help

Server Status

Applications					
Path	Version	Display Name	Running	Sessions	Comments
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy [Expire sessions] with idle > 30 /minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy [Expire sessions] with idle > 30 /minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy [Expire sessions] with idle > 30 /minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy [Expire sessions] with idle > 30 /minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy [Expire sessions] with idle > 30 /minutes

Deploy

Deploy directory or WAR file located on server

Context Path:

Version (for parallel deployments):

WAR Configuration file path:

WAR or Directory path:

Deploy

WAR file to deploy

Select WAR file to upload: [Choose File](#) / No file chosen

Deploy

Configuration

Re-read TLD configuration files

TLD load name (optional):

Re-read

Now create a new job and give name as per your project and select the pipeline which you are working

Test with the sample pipeline like Hello Word using the script

```
pipeline {  
    agent any  
  
    stages {  
        stage('Hello') {  
            steps {  
                echo 'Hello World'  
            }  
        }  
    }  
}
```

Click on build now and check the console output it shows success

```
Started by user naresh  
[Pipeline] Start of Pipeline  
[Pipeline] node  
Running on Jenkins in /var/lib/jenkins/workspace/Automation  
[Pipeline] {  
[Pipeline] stage  
[Pipeline] { (Hello)  
[Pipeline] echo  
Hello World  
[Pipeline] }  
[Pipeline] // stage  
[Pipeline] }  
[Pipeline] // node  
[Pipeline] End of Pipeline  
Finished: SUCCESS
```

Now we need to create automate pipeline to deploy the application on tomcat server

Create a New job with the name of Pipeline-Automation

Create a script using pipeline syntax select the hello world

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
1 pipeline {  
2   agent any  
3  
4   stages {  
5     stage('Hello') {  
6       steps {  
7         echo 'Hello World'  
8       }  
9     }  
10  }  
11 }  
12 |
```

Hello World

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Remove the existing script and create new as per the below, change your github repo with your project and user your credentials for tomcat

pipeline {

agent any

stages {

stage('checkout') {

steps {

echo 'git checkout stage'

git branch: 'main', url: 'https://github.com/devopstraininghub/mindcircuit17d.git'

}

}

stage('build') {

steps {

echo 'building with maven'

sh 'mvn clean install'

}

}


```

stage('deploy') {
    steps {
        echo 'deploying to tomcat'

        deploy adapters: [tomcat9(alternativeDeploymentContext: '', credentialsId: 'tomcat', path: '',
url: 'http://98.130.124.2:8081/']], contextPath: 'facebooapp', war: '**/*.war'

    }
}
}
}

```

Apply and save or you can click on save button directly

Click on build now it will success


Jenkins / Pipeline-Automation

 Status

 Changes

 Build Now

 Configure

 Delete Pipeline

 Stages

 Rename

 Pipeline Syntax

Pipeline-Automation

Permalinks

Builds > ...

Today

 #1 7:09 AM ▼

Click on green tick button to view console output

You can view complete stages of the pipeline

```
Started by user naresh
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/Pipeline-Automation
[Pipeline] {
[Pipeline] stage
[Pipeline] { (checkout)
[Pipeline] echo
git checkout stage
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/devopstraininghub/mindcircuit17d.git
> git init /var/lib/jenkins/workspace/Pipeline-Automation # timeout=10
Fetching upstream changes from https://github.com/devopstraininghub/mindcircuit17d.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/devopstraininghub/mindcircuit17d.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/devopstraininghub/mindcircuit17d.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 4da8aa9c1b6bb250a67eb4a0261f80d03179ea14 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 4da8aa9c1b6bb250a67eb4a0261f80d03179ea14 # timeout=10
> git branch -a -v --no-abbrev # timeout=10
> git checkout -b main 4da8aa9c1b6bb250a67eb4a0261f80d03179ea14 # timeout=10
Commit message: "Add Jenkins pipeline for checkout, build, and deploy"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (build)
[Pipeline] echo
building with maven
[Pipeline] sh
+ mvn clean install
[INFO] Scanning for projects...
[INFO] -----< com.example:facebookapp >-----
[INFO] Building facebookapp 1.0.0
[INFO] -----[ war ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ facebookapp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ facebookapp ---
```

```
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform
dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/Pipeline-Automation/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ facebookapp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ facebookapp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform
dependent!
[INFO] skip non existing resourceDirectory /var/lib/jenkins/workspace/Pipeline-Automation/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ facebookapp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ facebookapp ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.3.2:war (default-war) @ facebookapp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [facebookapp] in [/var/lib/jenkins/workspace/Pipeline-
Automation/target/facebookapp-1.0.0]
[INFO] Processing war project
[INFO] Copying webapp resources [/var/lib/jenkins/workspace/Pipeline-Automation/src/main/webapp]
[INFO] Building war: /var/lib/jenkins/workspace/Pipeline-Automation/target/facebookapp-1.0.0.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ facebookapp ---
[INFO] Installing /var/lib/jenkins/workspace/Pipeline-Automation/target/facebookapp-1.0.0.war to
/var/lib/jenkins/.m2/repository/com/example/facebookapp/1.0.0/facebookapp-1.0.0.war
[INFO] Installing /var/lib/jenkins/workspace/Pipeline-Automation/pom.xml to
/var/lib/jenkins/.m2/repository/com/example/facebookapp/1.0.0/facebookapp-1.0.0.pom
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.644 s
[INFO] Finished at: 2026-04-19T07:09:43Z
[INFO] -----
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (deploy)
[Pipeline] echo
deploying to tomcat
[Pipeline] deploy
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying /var/lib/jenkins/workspace/Pipeline-Automation/target/facebookapp-
1.0.0.war to container Tomcat 9.x Remote with context facebooapp
  Redeploying [/var/lib/jenkins/workspace/Pipeline-Automation/target/facebookapp-1.0.0.war]
  Undeploying [/var/lib/jenkins/workspace/Pipeline-Automation/target/facebookapp-1.0.0.war]
  Deploying [/var/lib/jenkins/workspace/Pipeline-Automation/target/facebookapp-1.0.0.war]
[Pipeline] }
[Pipeline] // stage
```

```
[Pipeline] }  
[Pipeline] // node  
[Pipeline] End of Pipeline  
Finished: SUCCESS
```

Now go to the tomcat GUI and refresh the page to view your deployment package file

Applications		
Path	Version	
/	None specified	Welcome
/docs	None specified	Tomcat D
/examples	None specified	Servlet a
/facebooapp	None specified	Archetyp
/host-manager	None specified	Tomcat H
/manager	None specified	Tomcat M

Deploy
Deploy directory or WAR file located on server

Open the facebook app on new tab you can see your application



Once you'r done, do not forget to stop or terminate instance to get any surprises for unused bill